Team Responsibilities and Milestones

Project: Measuring correlated failure in multi-agent visual evidence resolution **Status:** Proposed allocation for team and advisor review. Supersedes the work split in the superseded `PROJECT_EXECUTION_PLAN.md` (archived) if the framing in the accompanying `PROPOSAL_CORRELATED_FAILURE.md` is approved. **Date:** 2026-10-27 **Window:** 2026-10-27 to 2027-01-22

Roles

Member	Role	Owns
Aditya Hegde	Project lead	The measurement definitions, the deterministic core under concurrency, the evaluation, and every frozen contract. The research claim stands or falls on this track.
Rishi Sheth	Simulation and scenario generation	The controlled-correlation scenario corpus and the optional simulator spike.
Pranav Senghal	Harness, baselines, and release	The sweep infrastructure, comparison policies, and the artifacts that make the work usable and presentable.

Each member owns their track's correctness. The lead owns cross-track contracts and is the only person who changes a frozen definition.

Three parallel tracks

The tracks are designed to run concurrently after milestone M1. Only M1 is a hard serialization point, because every track consumes the contracts frozen there.

Track A - Measurement and core (Aditya)

The hardest and least delegable work. Every item here is either a definition the rest of the project depends on, or a concurrency problem with no obvious correct answer.

- **A1. Freeze the dependence measure.** Define the dependence inflation factor (observed joint error rate over independence-predicted joint error rate), its estimator, its sample-size requirements, and its failure cases. This is the project's central definition.
- **A2. Freeze the agreement-reliability measure.** Define P(correct | k sources agree) as a function of estimated dependence, including how it is computed without simulator truth at resolution time.
- **A3. Deterministic replay under out-of-order and late-arriving evidence.** Define and version the arrival policy: watermark, late-arrival window, and the behavior when a claim arrives after a trace event it would have changed. Prove it with property-based tests asserting that any arrival permutation within the watermark window yields an identical final belief and trace.
- **A4. Resolver and baseline set.** Implement an explicit, versioned dependence treatment, and reimplement the naive-independence rule (a differing `source_id` taken as statistical independence) as one of the comparison baselines. Sprint 1 is not frozen; the commitment is to the baseline's behaviour, not to the prototype's code.
- **A5. Evaluation and analysis.** Produce the correlation-versus-task-error characterization, the operating boundary, and the final written evidence.

Track B - Scenario generation and simulation (Rishi)

- **B1. Latent-variable scenario generator.** Generate claim sequences from a scene model where the degree and source of inter-agent correlation are set deliberately. Must support the four correlation channels: shared occluder, shared detector model version, shared viewpoint geometry, shared illumination.
- **B2. Corpus and seed management.** Fixed-seed, versioned scenario corpus with retained generation parameters, so every reported run is reproducible.
- **B3. Ground-truth separation.** Simulator truth is emitted alongside, never inside, the agent-visible claim stream, per AD-001. Rishi owns the mechanism that keeps them separate.
- **B4. PX4 and Gazebo two-X500 spike.** Off the critical path. Run it only after B1 to B3 are delivered. A failed spike must leave the core evaluation fully runnable.

Track C - Harness, baselines, and release (Pranav)

- **C1. Sweep harness.** Run the full scenario matrix across policies, retain every run, and emit comparable result tables. Must be re-runnable end to end by one command.

- **C2. Baseline policies.** Implement the comparison set: latest-report, first-report-wins, unweighted majority vote, and always-verify, each behind the existing `ResolutionPolicy` interface.
- **C3. Release engineering.** Root `README.md`, an Apache-2.0 `LICENSE`, an installable package, and a single quickstart command that works on a clean clone. Note that MultiUAV-Plat is GPL-3.0; no GPL components enter this core.
- **C4. Visual demonstration.** A rendered walkthrough of one correlated-failure scenario, showing two agents agreeing, being wrong together, and the trace explaining the resulting task decision.
- **C5. Integration documentation.** The camera-to-claim adapter contract, documented but implemented only if B4 passes.

Milestones

ID	Date	Deliverable	Owner	Gate
M0	Nov 3	Advisor approves the research framing	Aditya	Blocking. No downstream work starts without it.
M1	Nov 10	Contracts frozen: dependence measure, agreement-reliability measure, four correlation channels, scenario matrix, acceptance thresholds	Aditya	Blocking for B1 and C1. Tracks run in parallel after this.
M2	Nov 24	Generator produces controlled-correlation corpus (B1-B3); out-of-order determinism proven by property tests (A3); baselines implemented (C2)	All three, parallel	Determinism tests pass; corpus reproduces from seed.
M3	Dec 12	First end-to-end sweep. Initial correlation-versus-task-error curves	Aditya and Pranav	A real curve exists, even if ugly.
M4	Dec 18	Release artifacts complete (C3); PX4 spike go or no-go decision (B4)	Pranav, Rishi	Clean clone runs the demo. Spike result recorded either way.

ID	Date	Deliverable	Owner	Gate
M5	Jan 15	Final runs, analysis, operating boundary, visual demo (A5, C4)	All three	Every reported number traces to a retained run.
M6	Jan 22	Workbook and demo evidence submitted	All three	Stage 3 deadline.

Dependencies and handoffs

- **M1 is the only hard serialization.** Rishi cannot build the generator until the correlation channels are defined; Pranav cannot build the sweep harness until the metrics are defined. Aditya must not let M1 slip.
- **A3 and B1 are independent** and must run in parallel. Determinism work does not need the corpus, and the generator does not need the watermark policy.
- **C2 is independent of everything** after M1 and should start immediately, so Pranav is never blocked waiting on the lead.
- **Rishi hands Aditya** a corpus plus its generation parameters. Aditya never edits the corpus; defects go back to Rishi.
- **Pranav hands Aditya** result tables, never conclusions. Interpretation sits in Track A.
- **Nobody but Aditya changes a definition frozen at M1.** Those are the project's immutable contracts. Note that the Sprint 1 prototype is **not** among them: AD-005 was withdrawn on 2026-10-27 and that code, its fixtures, and its test assertions carry no protected status.

Risks and fallbacks

1. **M1 slips.** Highest-impact risk, because two tracks idle. Fallback: Aditya freezes a deliberately simple first version of the dependence measure and versions it, rather than waiting for the right one.
2. **The generator cannot produce clean correlation control.** Fallback: reduce from four correlation channels to two (shared occluder and shared model version), which still supports the central claim.

3. **The PX4 spike fails.** Already handled by design. It is off the critical path and the evaluation remains fully runnable without it.
4. **Correlation shows no measurable effect on task outcomes.** This is a publishable negative result, not a project failure, provided the sweep is reproducible and the measurement is sound. It must be reported, not buried.
5. **Uneven load late in the schedule.** Track A is heaviest from M3 onward. Pranav's C4 and C5 are the intended buffer and can absorb reallocation.

Weekly cadence

One team-only meeting per week and one advisor meeting every two weeks, per the course requirement. Each meeting record states ownership and action items and is kept separate from milestone updates.